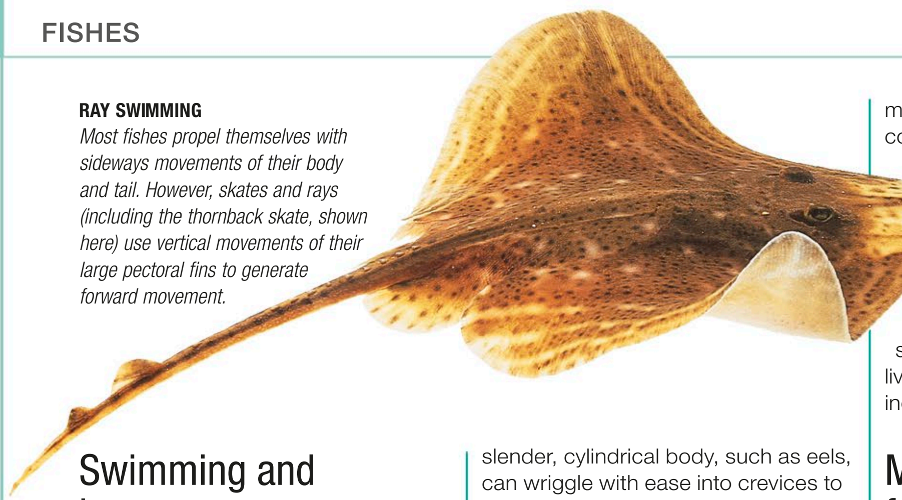


RAY SWIMMING
Most fishes propel themselves with sideways movements of their body and tail. However, skates and rays (including the thornback skate, shown here) use vertical movements of their large pectoral fins to generate forward movement.

Swimming and buoyancy

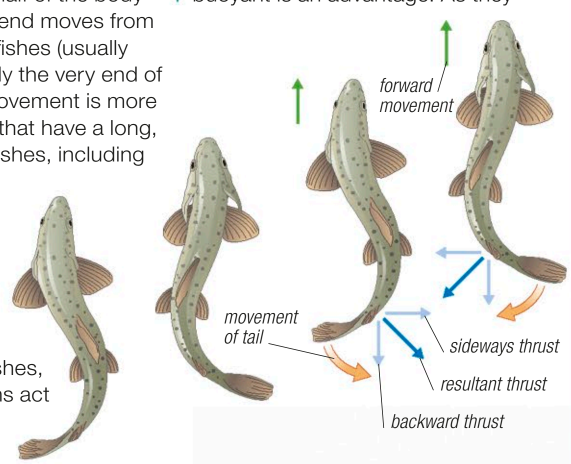
A fish propels itself through water using the action of its muscles. Along either side of the backbone are bundles of muscle fibers (known as myotomes). By contracting these muscle bundles in sequence, one a fraction of a second after the other, the fish creates a wave motion that travels along its body from front to back and ultimately causes the tail to move from side to side. It is this wave motion that propels the fish through the water. The wave usually begins near the rear third or half of the body so that only the back end moves from side to side. In some fishes (usually fast-moving ones), only the very end of the tail moves. The movement is more pronounced in fishes that have a long, slender body. Some fishes, including eels, can reverse the direction of the wave movement and so swim backward.

Fins play an important part in propulsion and movement. In most fishes, the dorsal and anal fins act in a similar way to the keel of a boat, stabilizing the fish's body. The paired fins have various uses including propulsion. Most fishes use them as control surfaces, adjusting the angle of the fins to move up and down in the water. In sharks and some ray-finned fishes, the paired fins also provide lift as the fish moves forward, while some fishes use their paired fins to "walk" on the bottom and, on rare occasions, on land. The tail fin is used mainly for propulsion but it also has a role in steering.

In general, fishes move relatively slowly, rarely exceeding 3 mph (5 kph). However, some fishes are capable of moving at much greater speed—for example, the wahoo (a close relative of tunas) can reach 50 mph (75 kph) for short periods.

A fish's body shape usually reflects the way it swims. Fishes that need to swim at high speeds in open water for long periods typically have a torpedo-shaped (fusiform) body. A body that is tall and flattened from side to side is less efficient but is common among reef fishes, which need to make sudden turns around dense vegetation or rocky surfaces. Fishes with a slender, cylindrical body, such as eels, can wriggle with ease into crevices to find food or escape from predators. Bottom-dwelling fishes typically have a body that is compressed from top to bottom, helping them remain inconspicuous on the seafloor.

Many fishes are neutrally buoyant. That is, their density is similar to that of the surrounding water, so they can remain suspended at a constant depth in the water column. Most other fishes are slightly negatively buoyant—their fins give them lift as they move, but they will sink if motionless. For bottom-dwelling fishes, such as skates and rays, being negatively buoyant is an advantage. As they move up and down in the water column, most fishes can adjust their buoyancy using a gas-filled organ known as the swim bladder (see p.487). To make themselves more or less buoyant, gas is added to or extracted from this bladder. Cartilaginous fishes do not have a swim bladder but their large, oil-rich liver is less dense than water and thus increases their buoyancy.



PROPULSION
As a fish's tail moves, it exerts a sideways thrust and a backward thrust on the water. The resultant force acts at an angle halfway between these 2. Movement of the tail from side to side means that the resultant thrusts to left and right produce a net backward thrust. Because every force in nature produces an equal and opposite reaction, this backward thrust propels the fish forward.

SURVIVING IN FRESH AND SALT WATER
The European eel can survive in both fresh and salt water. It spends most of its adult life in rivers but returns to sea to breed. The opposite is true of many salmon and trout, which move from the sea to fresh water to breed.

Marine and freshwater fishes

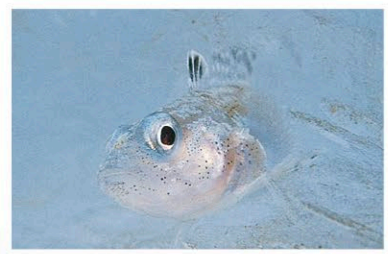
Fishes need mechanisms to keep the concentration of water and salts in their body in a stable state. However, this concentration is different from the water in which they live, so they have to counteract osmosis—the tendency of liquids of different concentrations (in their cells and in their environment) to equate by passing through a semipermeable membrane (the cell walls). Water always flows from a weaker solution to a more concentrated one to dilute it. Therefore marine fish tend to dehydrate and lose water because the salt concentration in their cells is lower than that of the surrounding seawater, and they must drink seawater (their bodies are adaptated to do this). In contrast, fish that live in fresh water have a tendency to take in water and must produce copious amounts of urine to compensate. Fish that move between salt and fresh water (such as salmon and eels) are physiologically able to adapt as long as they take it slowly. In fishes, water is involuntarily lost and gained through the mouth and gills. To prevent any excesses that could be detrimental to their internal chemistry, fishes "osmoregulate" in different ways.

In marine fishes, the concentration of salts is lower inside their bodies than in the water. Sharks, rays, and coelacanths prevent dehydration by retaining urea, which significantly increases their cell fluid concentration. Other marine fishes counteract water loss by drinking large quantities of seawater and excreting most of the salt. Some have chloride cells in their gills and on the inside the gill cover (operculum) that remove salts from the water that is absorbed and then excrete it. Freshwater fishes control involuntary water intake from the rivers and lakes in which they live by drinking little and producing copious dilute urine. They obtain the salt they need from their food and have cells in their gills that control its loss.

Temperature control

Fishes are ectothermic (cold-blooded) animals, which means that their internal temperature is the same as that of the surrounding water. Some species modify their behavior, basking in the sun, seek shade, or rise and fall between warmer and colder depths. Others do this by altering their color, using pigments that change between dark, heat-absorbing colors and light, heat-reflecting shades.

A few fishes—including tuna and the great white shark and its relatives—have some similarities with warm-blooded animals. Some of the heat generated by their large swimming muscles is conserved, so that their internal temperature is higher than that of the water around them. This makes it possible for these fishes to remain active in cold water.



SURVIVING COLD
Seawater freezes at a lower temperature than fish blood. However, the blood of some fishes, such as this icefish, remains liquid even in extremely cold water because it contains proteins that act like antifreeze.